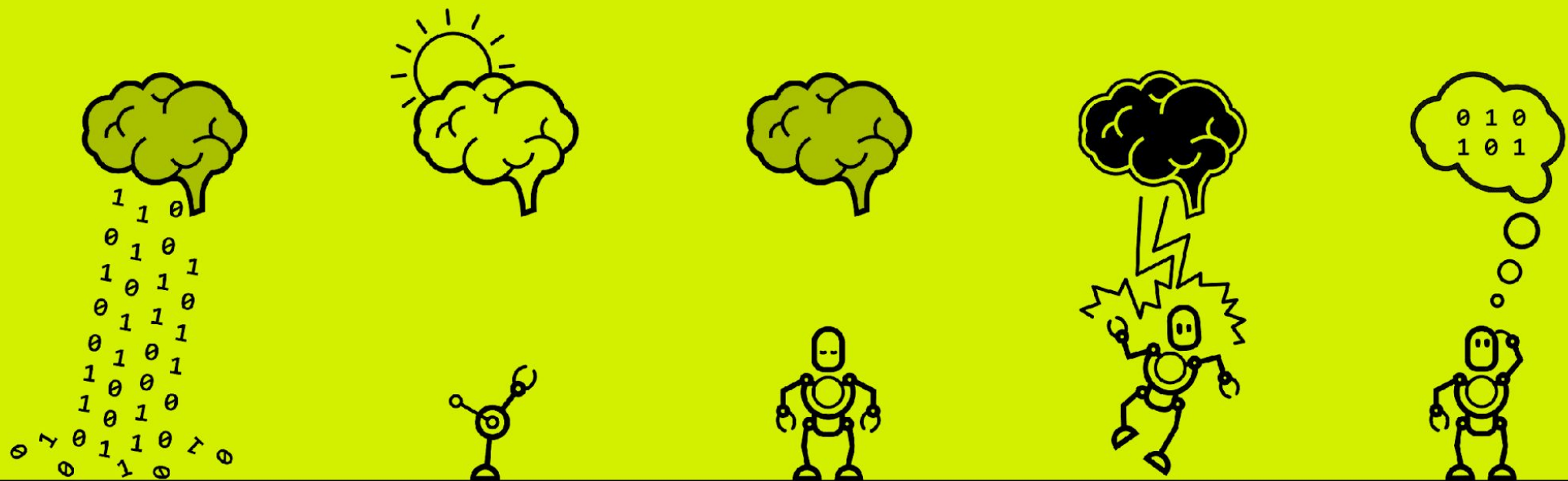




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THE BRAIN-COMPUTER INTERFACE
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NeuroVision: ECoG-Based Visual Decoding

Makar Lavrov – Tobías Fleitas
[Github Repository](#)

Introduction

Understanding Visual Processing in the Human Brain

- **Data:**

ECoG recordings from an epilepsy patient watching a video

- **Brain regions:**

Temporal & visual cortex

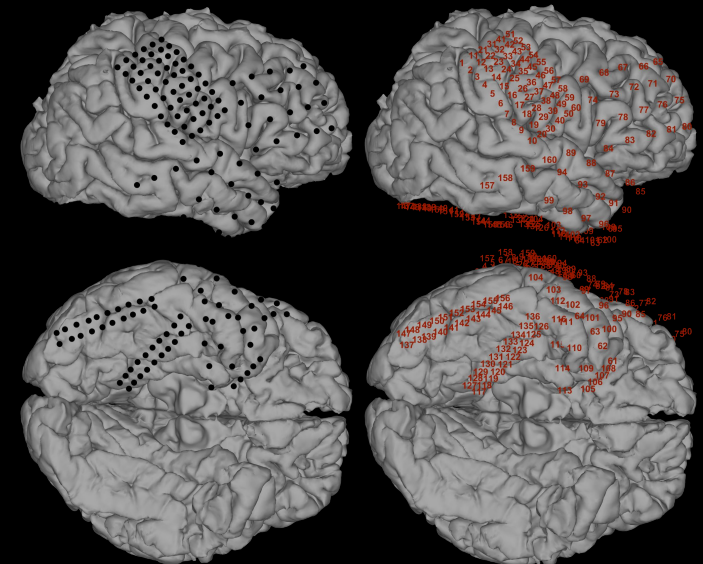
- **Task:**

Classify what the subject sees:

→ Color | Shape | Face

- **Side Objective:**

Compare decoding performance: ECoG vs EEG



Idea

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Raw Signal ECoG

Preprocessing

Feature
Extraction

Classification

Prediction

Clinical Matrix

100 ROI channels
(61–160) from temporal
base; 1200 Hz
sampling.

Noise Suppression

Zero-phase Notch
(50/100/150 Hz) + CAR
(Common Average
Reference)

Neural Biomarkers

Hilbert-based
High-Gamma envelope
(70–150 Hz); Mean
power epochs.

ML Optimization

Random Forest vs.
SVM (RBF); 5-fold
cross-validation.

Visual Decoding

Real-time classification
of Color, Shape, and
Face stimuli.

Implementation

- **Data Processing**

- MATLAB .mat ECoG dataset
- 100 selected ROI channels
- 126 stimulus events (epochs)

- **Signal Processing**

- Sampling rate: 1200 Hz
- Epoch window: -0.2s to 1.0s
- High-Gamma extraction (70–150 Hz)

- **Feature Matrix**

- Features: mean power per channel
- Final shape: (126 × 100)

- **Machine Learning**

- Models: Random Forest, SVM
- Evaluation: 5-fold cross-validation

Results

• Classification Accuracy (5-fold Cross-Validation)

Random Forest: 66.55% ($\pm 9.56\%$)
 SVM (RBF Kernel): 61.08% ($\pm 4.88\%$)

• State-of-the-Art (SOTA) Alignment

Our 66.55% accuracy for dynamic video stimuli is highly competitive with current benchmarks (typically 65–75%).

• Direct Comparison: ECoG vs. Unicorn (EEG)

ECoG (Intracranial): 66.55%
 Unicorn (Scalp): 43.70%



Reflection

Signal Quality & Modality

ECoG Advantage: Direct cortical recording captures High-Gamma (70–150 Hz), which is essential for visual decoding but nearly invisible to scalp EEG.

Lesson Learned

Bandwidth: While EEG is restricted to lower frequencies (Alpha/Beta), ECoG's broadband signal is necessary for high-accuracy BCI.

Optimization & Hardware

Memory Management: Built an "OOM-safe" pipeline to process 160-channel clinical data on an 8 GB RAM system.

Group Picture



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ECoG Video Analysis

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